

our related work discussion on systems for logic programming beyond SQL. We relate Lobster to works along three directions: accelerated Datalog engines, probabilistic and differentiable programming, and neurosymbolic methods.

High-Performance Datalog A variety of Datalog-based systems have been built for program analysis [14, 21, 36] and even enterprise database applications [3], though these systems run exclusively on the CPU. The FVLog system [38, 44] provides a Datalog engine implemented for GPUs, but it lacks support for the probabilistic and differentiable reasoning needed for deep learning integration. Moreover, FVLog focuses on the domain of large analytics queries, which emphasizes simpler queries executed against large databases, which is not a focus for Lobster. Notably, FVLog does not come with a query planner and user-facing front-end, meaning that users need to directly interact with low-level relational algebra operations supported by the system.

Probabilistic and Differentiable Programming *Probabilistic programming* allows programmers to model distributions and perform probabilistic sampling and inference [5, 13, 16, 46]. *Differentiable programming* systems allow programmers to write code that is differentiable and therefore amenable to use during neural network training. Symbolic and automatic differentiation [4] are commonly used in popular ML frameworks such as PyTorch and others [1, 15, 33].

Probabilistic programs are not in general differentiable and thus cannot be run during training. The differentiable programming systems described above are designed for real-valued functions and are not compatible with logic programming. Lobster, on the contrary, focuses on the differentiability of logic programs with probabilities.

Neurosymbolic Methods The emerging domain of neurosymbolic computation combines symbolic reasoning into existing data-driven learning systems. There have been a large number of successful neurosymbolic systems across a range of machine learning domains like computer vision and natural language processing [7–11, 24, 28–32, 37, 41, 47, 49–53]. Lobster builds upon the Scallop neurosymbolic programming language [20, 26], as Scallop is general enough to implement other neurosymbolic systems [8, 30, 49, 50]. However, Lobster improves upon the CPU-only Scallop by using GPU acceleration to provide higher performance and the ability to scale to larger datasets, as we demonstrated in Section 6.

8 Conclusion

We have described the design and implementation of the Lobster neurosymbolic engine. With existing engines, symbolic computation can quickly become the bottleneck when neural computations benefit from domain-specific hardware accelerators like GPUs. Lobster shows how Datalog programs can also take advantage of GPUs, providing large speedups and strong scalability over CPU-only engines like Scallop.

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Algorithm 1: How to execute a RAM program via compilation to and execution of APM.

Data: Database F_T , RAM program $\bar{\phi}$.

Result: F_T updated to reflect the result of evaluating $\bar{\phi}$.

```

1 for  $\phi$  in  $\bar{\phi}$  do
2    $instructions \leftarrow compile(\phi)$ ;
3    $F_T^{stable}, F_T^{recent}, F_T^\Delta \leftarrow \emptyset, F_T, \emptyset$ ;
4    $size \leftarrow |F_T^{stable}|$ ;
5   while true do
6     for  $i$  in  $instructions$  do
7        $\lfloor$  execute  $i$ 
8      $size_{new} \leftarrow |F_T|$ ;
9     if  $size_{new} = size$  then
10       $\lfloor$  break;
11     $size \leftarrow size_{new}$ ;
12   $F_T \leftarrow F_T^{stable}$ 

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A RAM to APM Translation

We detail the compile function used to translate RAM to APM in Figure 14.

B APM Evaluation

Details on executing a RAM program via compilation to APM are provided in Algorithm 1.